

## **SCOPE OF WORK**

# **TTS feasibility and (pre)-/clinical supply Aminophylline TTS**

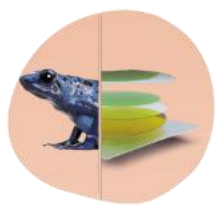
**Authors:** Dr. C. Lohmann, Dr. J. Lodder-Gadaczek, Dr. P. Mohr, Dr. D. Pirwitz  
Bernardi

LTS Lohmann Therapie-Systeme AG, Germany

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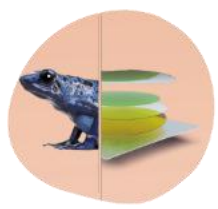
**Project type:** TTS Feasibility

**Date:** May 21, 2026



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## Introduction

The Johannes Gutenberg University Mainz wishes to have LTS develop a transdermal formulation of Aminophylline to show the preclinical and clinical feasibility of this API following the transdermal route of administration.

## Target Product Profile

| Parameter               | Properties & product design                                                                                                     |
|-------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| API                     | Aminophylline                                                                                                                   |
| Dosage Form             | Transdermal Therapeutic System                                                                                                  |
| Indication              | Multiple Sclerosis                                                                                                              |
| Target Dose             | ~ 300 mg / ~ 60 cm <sup>2</sup> / 4 days (based on preliminary permeation results)<br><i>In vitro</i> : 50 µg/cm <sup>2</sup> h |
| Route of administration | transdermal                                                                                                                     |
| Application time        | 4-day patch                                                                                                                     |
| Project Scope           | Provision of preclinical samples to evaluate skin irritation and plasma levels over 4 days                                      |

## Background Information

Based on the positive outcome of the initial pre-formulation development of the Johannes Gutenberg University Mainz using Aminophylline in a transdermal system, optimizations in terms of adhesion, manufacturability and process knowledge will be considered.

## Goals and Objectives

List the goals and objectives for this project:

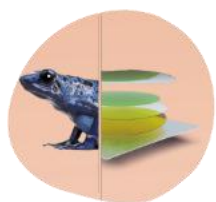
- Development of prototypes, supported by in-vitro skin permeation trials
- Provision of preclinical samples for a preclinical animal study
- Provision of clinical samples for an exploratory clinical trial

## Overall deliverables

- Regular meetings/telcons to exchange on project progress, e.g. project updates in presentation style
- Written reports on agreed milestones
- Required data documentation

## Assumptions

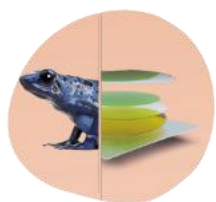
- API and reference standards and other materials will be provided by LTS
- API is stable and no critical instabilities known
- No additional equipment needed for laboratory activities
- Focus will be on the enhancer previously selected by the Johannes Gutenberg University Mainz
- Preclinical and clinical study will be performed under responsibility of the partner. Timelines for such trials need to be defined
- No change of formulation will be needed after availability of the preclinical outcome



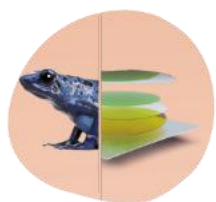
## Scope of Work & Deliverables

The following table lists the activities as outlines in the Gantt chart. The work package includes all excipients and lab supplies. The required API and reference standards are included in the end price.

| WP | Activities / Deliverables of Working Package                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Estimated Time | Costs & payment schedule |
|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|--------------------------|
| 1  | <b>Prototype development including skin permeation assessment</b><br><br><u>Goal:</u><br>Development of suitable prototypes to deliver the API according to the QTPP and to select candidates for a preclinical study<br><br><u>Activities:</u> <ul style="list-style-type: none"><li>• Development of an analytical method to quantify API in Franz cell acceptor medium --&gt; already available</li><li>• Evaluation of adhesives and excipients</li><li>• Development of first formulation prototypes using standard LTS technologies</li><li>• Comparative in-vitro skin permeation testing of formulation prototypes in vitro skin permeation on skin tissue (up to 3 experiment cycles on human or minipig skin)</li></ul><br><u>Deliverables:</u><br>Reporting and discussion of permeation results (flux and cumulative data), incl. selection of up to 2 suitable prototypes for preclinical study | ~3.5 months    | 136.000€                 |
| 2  | <b>Manufacturing of preclinical samples</b><br><br><u>Goal:</u><br>To have non-GMP samples ready for a pre-clinical trial<br><br><ul style="list-style-type: none"><li>• Development of an ID and assay method for selected prototypes</li><li>• Manufacture of lab scale non-GMP samples (2 formulations, 1 size) for pre-clinical trial, incl. release (ID/assay)</li></ul><br>Note: Pseudo-stability of samples to be assessed at supplementary samples returning from preclinical trial<br><br><u>Deliverables:</u><br>Samples for pre-clinical trial ready for shipment (typically ~50 TTS per formulation)                                                                                                                                                                                                                                                                                             | ~1.5 months    | 96.336€                  |
| 3  | <b><u>Development of analytical methods / stability studies</u></b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | ~6 months      | 402.000€                 |



|                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                   |            |    |    |     |     |     |     |     |            |   |  |   |   |   |   |  |  |            |   |   |   |  |  |  |  |                                        |  |
|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|------------|----|----|-----|-----|-----|-----|-----|------------|---|--|---|---|---|---|--|--|------------|---|---|---|--|--|--|--|----------------------------------------|--|
|                              | <p><u>Goal:</u><br/>Development of suitable analytical methods</p> <ul style="list-style-type: none"><li>Manufacturing of samples for method development and stability studies</li><li>Development of analytical methods (assay (incl. partial validation appropriate for an exploratory study), assay of 1 functional excipient, in-vitro dissolution, degradation, ID, peel adhesion, release liner peel, cold flow, swab)</li><li>Short term stability study (1 formulation, 1 size)</li></ul> <p>Conditions:</p> <table><tr><td></td><td>T0</td><td>T1</td><td>T3</td><td>T6</td><td>T9</td><td>T12</td><td>T18</td><td>T24</td></tr><tr><td>25°C/60%RH</td><td rowspan="2">X</td><td></td><td>X</td><td>X</td><td></td><td></td><td></td><td></td></tr><tr><td>40°C/75%RH</td><td>X</td><td>X</td><td>X</td><td></td><td></td><td></td><td></td></tr></table> <p><u>Deliverables:</u><br/>Analytical methods and first stability outcomes are available</p>                                                                                                                                                                     |                                                                   | T0         | T1 | T3 | T6  | T9  | T12 | T18 | T24 | 25°C/60%RH | X |  | X | X |   |   |  |  | 40°C/75%RH | X | X | X |  |  |  |  |                                        |  |
|                              | T0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | T1                                                                | T3         | T6 | T9 | T12 | T18 | T24 |     |     |            |   |  |   |   |   |   |  |  |            |   |   |   |  |  |  |  |                                        |  |
| 25°C/60%RH                   | X                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                   | X          | X  |    |     |     |     |     |     |            |   |  |   |   |   |   |  |  |            |   |   |   |  |  |  |  |                                        |  |
| 40°C/75%RH                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | X                                                                 | X          | X  |    |     |     |     |     |     |            |   |  |   |   |   |   |  |  |            |   |   |   |  |  |  |  |                                        |  |
| 4                            | <p><b><u>Preparation of clinical trial supply</u></b></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | ~6 months                                                         | 206.000€   |    |    |     |     |     |     |     |            |   |  |   |   |   |   |  |  |            |   |   |   |  |  |  |  |                                        |  |
|                              | <p><u>Goal:</u><br/>To have GMP samples ready for a clinical trial</p> <ul style="list-style-type: none"><li>Establishing of material specification of excipients &amp; API</li><li>Release testing of API and starting materials</li><li>Compilation of CMC documents</li><li>Documentation / organization of GMP manufacturing</li><li>GMP manufacture on GMP lab scale of exploratory clinical trial material</li><li>1 active prototype batch (1 size/strength) including GMP documentation</li><li>Release testing of samples and QP release of phase I clinical trial material</li><li>Starting a stability program of the clinical trial batch</li></ul> <p>Conditions:</p> <table><tr><td></td><td>T0</td><td>T1</td><td>T3</td><td>T6</td><td>T9</td><td>T12</td><td>T18</td><td>T24</td></tr><tr><td>25°C/60%RH</td><td rowspan="2">X</td><td></td><td>X</td><td>X</td><td>X</td><td>X</td><td></td><td></td></tr><tr><td>40°C/75%RH</td><td>X</td><td>X</td><td>X</td><td></td><td></td><td></td><td></td></tr></table> <p><u>Deliverables:</u><br/>Samples for clinical trial ready for shipment (typically ~50 TTS)</p> |                                                                   | T0         | T1 | T3 | T6  | T9  | T12 | T18 | T24 | 25°C/60%RH | X |  | X | X | X | X |  |  | 40°C/75%RH | X | X | X |  |  |  |  | WP4 to be started overlapping with WP3 |  |
|                              | T0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | T1                                                                | T3         | T6 | T9 | T12 | T18 | T24 |     |     |            |   |  |   |   |   |   |  |  |            |   |   |   |  |  |  |  |                                        |  |
| 25°C/60%RH                   | X                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                   | X          | X  | X  | X   |     |     |     |     |            |   |  |   |   |   |   |  |  |            |   |   |   |  |  |  |  |                                        |  |
| 40°C/75%RH                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | X                                                                 | X          | X  |    |     |     |     |     |     |            |   |  |   |   |   |   |  |  |            |   |   |   |  |  |  |  |                                        |  |
| Costs (incl. material costs) |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Overall timeline: ~15 months (until shipment of clinical samples) | 840.336€   |    |    |     |     |     |     |     |            |   |  |   |   |   |   |  |  |            |   |   |   |  |  |  |  |                                        |  |
| Total costs (incl. 19% VAT)  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                   | 1.000.000€ |    |    |     |     |     |     |     |            |   |  |   |   |   |   |  |  |            |   |   |   |  |  |  |  |                                        |  |



## Gantt Chart

